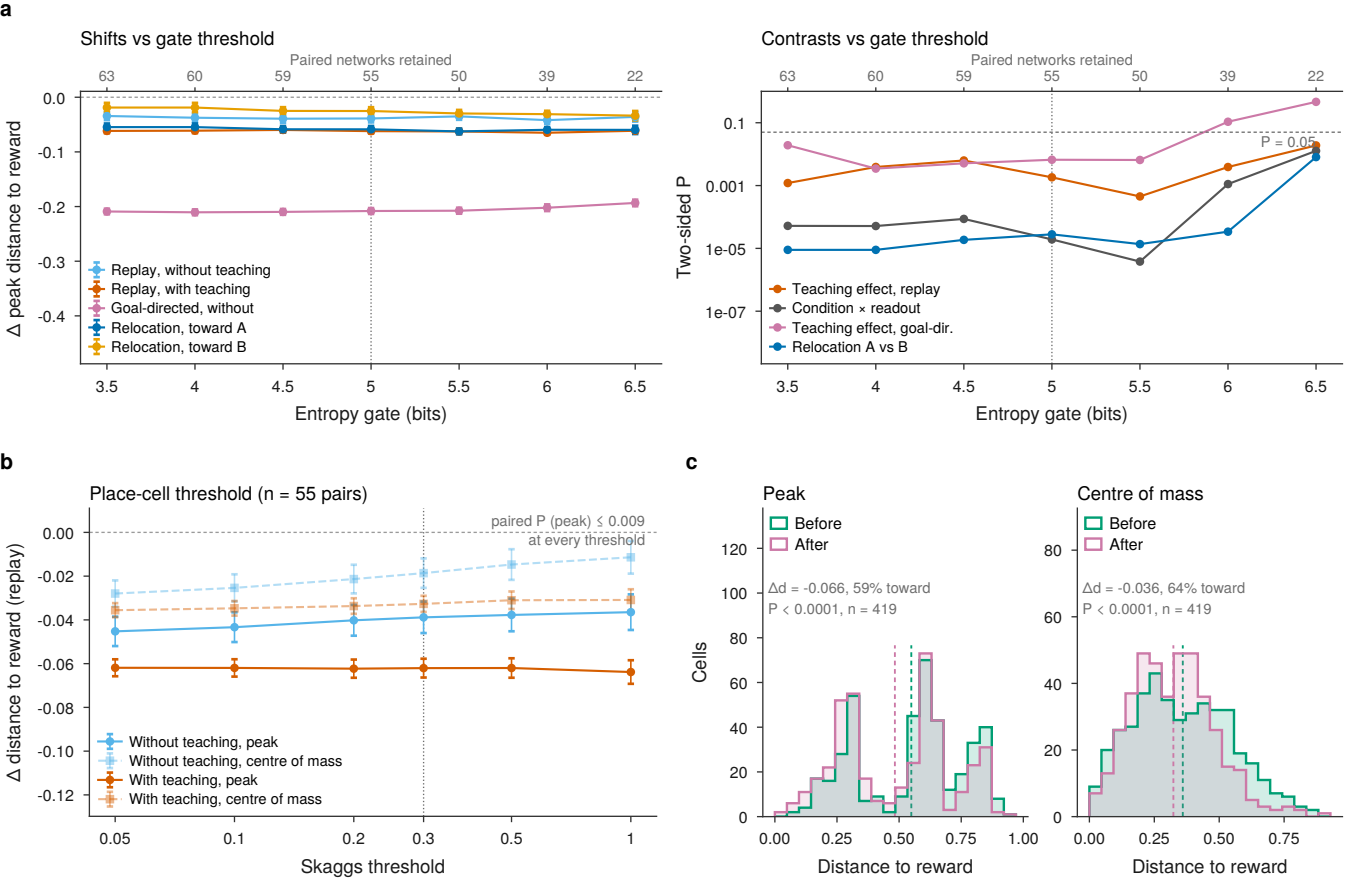

a
Shifts vs gate threshold
Paired networks retained
63
60
59
55
50
39
22
0.0
-0.1
-0.2
-0.3
-0.4
Δ peak distance to reward
Replay, without teaching
Replay, with teaching
Goal-directed, without
Relocation, toward A
Relocation, toward B
3.5
4
4.5
5
5.5
6
6.5
Entropy gate (bits)
Contrasts vs gate threshold
Paired networks retained
63
60
59
55
50
39
22
0.1
0.001
1e-05
1e-07
Two-sided P
P = 0.05
Teaching effect, replay
Condition × readout
Teaching effect, goal-dir.
Relocation A vs B
3.5
4
4.5
5
5.5
6
6.5
Entropy gate (bits)
b
Place-cell threshold (n = 55 pairs)
paired P (peak) ≤ 0.009
at every threshold
0.00
-0.02
-0.04
-0.06
-0.08
-0.10
-0.12
Δ distance to reward (replay)
Without teaching, peak
Without teaching, centre of mass
With teaching, peak
With teaching, centre of mass
0.05
0.1
0.2
0.3
0.5
1
Skaggs threshold
c
Peak
Before
After
Δd = -0.066, 59% toward
P < 0.0001, n = 419
120
100
80
60
40
20
0
Cells
0.00
0.25
0.50
0.75
1.00
Distance to reward
Centre of mass
Before
After
Δd = -0.036, 64% toward
P < 0.0001, n = 419
80
60
40
20
0
0.00
0.25
0.50
0.75
Distance to reward